

Technology #13: Self-Referencing Piezoresistive Pressure Sensor

Technology Domain: Piezoresistive Sensing + Tubing Mechanics + Drift Compensation

Mechanism: Catheter-integrated piezoresistive pressure sensor with self-referencing architecture for drift compensation. R1 fix: original P-27 suffered from chronic drift; R1 redesign adds self-referencing bridge for drift correction.

Maturity: ENGINEERING_DEFINITION

Current State: No physical prototype. Computational model and engineering definition exist.

Key Risk: Self-referencing cannot reduce drift below 1 mmHg/month (insufficient for clinical utility)

Next Experiment: MEMS sensor prototypes (various diaphragm thicknesses)

Buyer Receives: Self-referencing sensor concept + R1 correction (drift compensation architecture); Piezoresistive + Wheatstone bridge governing equations; Bench test protocols (calibration, drift, hermeticity, kink, EMC); External precedent catalog (MEMS pressure sensors, drift compensation); Critical UNKNOWN disclosure (achievable drift, hermeticity, sensor survival); This engineering dossier with failure analysis + build plan

Buyer Must Build: Production MEMS sensor design (diaphragm, piezoresistors, geometry); Self-referencing bridge architecture + signal conditioning; Hermetic packaging process; Catheter integration manufacturing; Regulatory submission (active implantable, ISO 14708-1, IEC 60601-1-2); Clinical validation (accuracy + drift over implantation duration); Long-term reliability testing

Transfer Posture: Sponsored Validation

Target Buyer: Implantable sensor company; MEMS manufacturer

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